Cellular Intelligence

1. scGPT & scDiffusion

scGPT: Towards Building a Foundation Model for Single-Cell Multi-omics Using Generative AI | bioRxiv

https://academic.oup.com/bioinformatics/article/40/9/btae518/7738782#480177449

- scGPT is transformer (with decoder autoregressive training, although hard to see)
    - tokenization, attention, ...
- scDiffusion using an encoder / decoder scheme (using scimilarity)
- scimilarity is a 4-L NN (each gene is a scalar value), 128 hidden
    - always 20K genes, had to retrain it because not the same set on their end
    - only on humans
    - trained with contrastive learning on triplet (positive/negative) and reconstruction
    - denoising too (40% dropout in input and 50% in model!)
    - hypersphere reg of the hidden emb
      → THIS IS NOT GAUSSIAN
- scDiffusion applies the regular diffusion scheme
    - classifier guided diffusion (cell type / organ / dev stages / ...)
        - could have been done better though
    - latent diffusion
- scDiffusion is not a foundation model
    - no fine tuning to dozens of novel tasks
    - only trained on 5 datasets

Diffusion

$$q(x_{\mathrm{i}} \,|\, \mathbf{x}_{i-1}) = N\left(x_{\mathrm{i}} \,\middle|\, \sqrt{1-\beta_i} x_{i-1}, \beta_i I\right), \approx \text{where} \quad \beta_i \in (0{,}1)$$

learn the reverse process p: predict the reverse noise

x_{i-1} = x_i - f(x_i, i)

$$p_\theta(x_{i-1}|x_i) = N\left(x_{i-1} \middle| \mu_\theta(x_i, i),\ \exp\left(w\beta_i\right) I\right)$$

$$p_\theta\left(x_{i-1} \mid x_i, y\right)$$
$$= \mathcal{N}\left(x_{i-1} \mid \mu_\theta\left(x_i, I\right) + \boxed{\beta_i \gamma \nabla_{x_i} \log p_\phi\left(y \mid x_i\right)}, \exp\left(w\beta_i\right) I\right)$$

*will use multiple such classifier (sums)

guidance

gradients of the log prob →

gradient interpolation (basically going from one gradient to another for 2 classes that you want to interpolate between using the gama1 and gama2

$$\rightarrow \beta_i \left(\gamma_1 \nabla_{x_i} \log p_\phi\left(y_1 \mid x_i\right) + \gamma_2 \nabla_{x_i} \log p_\phi\left(y_2 | x_i\right)\right)$$

Remarks

1. remarks in the paper the equations are often ill defined...
2. uses DDPM but no-one uses it.. it has always been DDIM (SDE→ ODE)

$$x_{t-1} = x_t + a_t \epsilon_\theta(x_t, \sigma_t) + \eta w_t$$

$$x_{t-1} = x_t + b_t \epsilon_\theta(x_t, \sigma_t)$$

capabilities

- can do generation with scDiff (harder with scGPT)
    - adds gradient interpolation to generate in between states
- can do denoising with scDiff (harder with scGPT)
- harder explainability of scDiff
- scDiff is more comp intensive
- scDiff is likely more brittle (2-stage; 2-model training) + classifier

→ **for your direct application scDiffusion is much more ready and aligned for what you want to do**

2. Exercise

predict effect of perturbations on cells

questions:

1. How did you get matched cells across time?
   → fake data I guess

setup

1. used codex to do a first version
2. iterate with codex

improvement in real-life

1. I would also redefine what are accurate losses and metrics in our context (see the papers from Maciej and from Shift Bio)
2. I would start with something like the LPM model (generating embeddings of molecules using
3. I would continue by merging LPM and scDiffusion further
4. I would also improve scDiffusion with what I mentioned above

→ But what you want to do is predict a set of perturbation to reach a specific cellular state

So I made a notebook to predict the perturbation given a required cell state (knowing that it should be a classification task but seeing it a regression task in this context) - it could become a soft matching or be paired with a a generative model downstream

Time list

1. part 1: 1h
2. part 2: 1h